

Financial Literacy Guide

S&P; 500 Outperformer Prediction Project — Group 15
Lana Gidan · Matthew Golubow · Shahd Tarman
Binghamton University · SSIE Department · 2025

This document explains every term, ratio, statistical measure, and concept used in this project in plain, accessible language. No prior finance or statistics background is assumed. Each term is defined, explained intuitively, and connected to what it means for this project specifically.

PART 1 — CORE CONCEPTS

1.1 The S&P; 500

The S&P; 500 is a list of 500 of the largest publicly traded companies in the United States — companies like Apple, ExxonMobil, Walmart, and JPMorgan. It is updated periodically as companies grow or shrink. Investors treat it as a proxy for the overall stock market. If the S&P; 500 goes up 5%, it means the average large American company gained 5% in value.

1.2 SPY — The Benchmark

SPY is an exchange-traded fund (ETF) that holds all 500 stocks in the S&P; 500 index. When someone says the market returned 7% this quarter, they usually mean SPY returned 7%. In this project SPY is our benchmark — the baseline every stock is compared against. If Apple returned 12% while SPY returned 7%, Apple beat the benchmark by 5%.

1.3 Outperformer vs Underperformer

These are the two categories your model predicts:

Outperformer (label = 1): The stock's quarterly return was GREATER than SPY's return for the same quarter.

Underperformer (label = 0): The stock's quarterly return was LESS than SPY's return for the same quarter.

Example: Apple returned +27.2% in Q3 2020. SPY returned +9.0% that quarter. Apple outperformed → label = 1.

1.4 Quarterly Return

The percentage gain or loss of a stock's price over one quarter (3 months).

Formula: $(\text{Price at end of quarter} - \text{Price at start of quarter}) \div \text{Price at start of quarter} \times 100$

Example: Apple was \$88.39 at start of Q3 2020 and \$112.45 at end. Return = $(112.45 - 88.39) / 88.39 = +27.2\%$.

1.5 Portfolio

A portfolio is a collection of stocks you own at the same time. Instead of putting all your money into one company, you spread it across several. In this project the portfolio is built from stocks your model predicts will outperform in the upcoming quarter. Each selected stock receives an equal share of the investment.

1.6 Portfolio Backtesting

Backtesting answers: if I had used this model in the past, would I have made money? You go back in time — using old historical data — and pretend to invest using the model's predictions. Then you check whether that pretend investment would have beaten just buying SPY. It is the proof that the model has real-world value, not just statistical value.

1.7 GICS Sectors

GICS (Global Industry Classification Standard) groups all companies into 11 industry sectors based on their primary business activity.

Term	Plain-English Meaning
Energy	Oil, gas, and energy companies (ExxonMobil, Chevron)
Information Technology	Software, hardware, semiconductors (Apple, Microsoft, NVIDIA)
Health Care	Pharma, biotech, medical devices (J&J, Pfizer)
Financials	Banks, insurance, investment firms (JPMorgan, Goldman Sachs)

Consumer Discretionary	Non-essential goods — cars, retail (Amazon, Tesla)
Consumer Staples	Essential everyday goods — food, beverages (Walmart, Costco)
Industrials	Manufacturing, aerospace, machinery (Caterpillar, Boeing)
Materials	Chemicals, mining, metals (Dow, Newmont)
Real Estate	REITs — property investment trusts (Prologis, Simon Property)
Utilities	Electric, water, gas utilities (NextEra, Duke Energy)
Communication Services	Telecom, media, internet (Alphabet, Meta, Verizon)

PART 2 — THE FINANCIAL RATIO PREDICTORS

These are the 13 variables fed into the logistic regression model. Each is calculated from a company's quarterly financial statements and captures a different aspect of the company's financial health.

Gross Margin

Formula: Gross Profit ÷ Revenue

What it measures: Of every \$1 of sales revenue, how many cents remain after paying the cost of making the product? A gross margin of 60% means the company keeps \$0.60 of every dollar before paying operating expenses.

In this project: High gross margin signals pricing power and efficient production. Tech companies typically have very high gross margins (60-80%). Retailers have thin margins (20-30%).

Operating Margin

Formula: Operating Income ÷ Revenue

What it measures: Of every \$1 of sales, how many cents remain after paying ALL operating costs (salaries, rent, marketing, R&D;) but before paying taxes and interest?

In this project: A shrinking operating margin quarter over quarter signals the company is becoming less efficient — a warning sign for investors.

Net Margin

Formula: Net Income ÷ Revenue

What it measures: The bottom line. Of every \$1 of sales, how many cents does the company keep as profit after paying everything — costs, salaries, taxes, and interest?

In this project: This is the most complete profitability measure. A 20% net margin means the company keeps \$0.20 of every dollar it earns.

ROA — Return on Assets

Formula: Net Income ÷ Total Assets

What it measures: How efficiently is the company using its assets (buildings, equipment, inventory, cash) to generate profit? A ROA of 10% means every \$100 of assets generates \$10 of profit.

In this project: Particularly important in Energy — oil companies own massive expensive assets. A high ROA means they use those assets well. Found significant in your Energy sector model.

ROE — Return on Equity

Formula: Net Income ÷ Total Equity

What it measures: How much profit does the company generate relative to the money shareholders have invested? Equity is what the company is worth after subtracting all debts.

In this project: Key in Real Estate — your analysis found ROE is the strongest predictor of outperformance among REIT companies. Found significant in your Real Estate sector model.

Current Ratio

Formula: Total Current Assets ÷ Total Current Liabilities

What it measures: Can the company pay its short-term bills (due within 12 months) using its short-term resources (cash, receivables, inventory)? Above 1.0 means yes. Below 1.0 signals trouble.

In this project: A current ratio of 2.0 means the company has \$2 of short-term assets for every \$1 of short-term debt — a comfortable buffer.

Debt-to-Equity

Formula: Total Liabilities ÷ Total Equity

What it measures: How much debt is the company carrying relative to what shareholders own? A ratio of 1.5 means \$1.50 of debt for every \$1 of equity.

In this project: Context matters. Banks and utilities normally carry high debt ratios — that is their business model. Tech companies with high debt are a warning sign. Never compare across sectors.

Asset Turnover

Formula: Revenue ÷ Total Assets

What it measures: How efficiently does the company use its assets to generate sales? A ratio of 0.8 means every \$1 of assets generates \$0.80 of revenue per quarter.

In this project: Critical in Energy. Companies that generate more revenue per dollar of assets tend to outperform. Found to be a significant predictor in your Energy sector model.

FCF Margin

Formula: (Operating Cash Flow + Capital Expenditures) ÷ Revenue

What it measures: Free cash flow is the actual cash left after the company pays for operations AND capital investments. FCF margin shows this as a percentage of revenue.

In this project: Profit can be manipulated by accounting choices. Free cash flow is harder to fake — it is actual cash in the bank. High FCF margin companies can fund growth, pay dividends, or buy back stock.

Cash Ratio

Formula: Cash & Equivalents ÷ Total Current Liabilities

What it measures: The strictest liquidity measure. Can the company pay all its short-term debts using only cash on hand — without selling anything?

In this project: Found to be the only significant predictor in the global model ($p = 0.040$). Companies with strong cash positions relative to obligations are more resilient in downturns.

Revenue Growth

Formula: Quarter-over-Quarter % change in Revenue

What it measures: Is the company growing its sales compared to the previous quarter? Positive = growing. Negative = shrinking.

In this project: A momentum signal — companies with accelerating revenue growth tend to attract investor attention and continue outperforming.

Net Income Growth

Formula: *Quarter-over-Quarter % change in Net Income*

What it measures: Is the company's profit growing compared to the previous quarter?

In this project: Found to be the strongest individual predictor in Consumer Staples. Earnings momentum is a well-established factor in finance research.

CapEx Ratio

Formula: *|Capital Expenditures| ÷ Revenue*

What it measures: What percentage of revenue does the company reinvest in physical assets — factories, equipment, technology infrastructure?

In this project: High CapEx can mean investing for future growth (positive) OR requiring constant heavy investment just to maintain operations (negative). Sector context matters greatly.

PART 3 — STATISTICAL MEASURES

p-value

The probability that your result happened purely by chance, assuming the model has no real predictive power. A p-value of 0.03 means there is only a 3% chance you got this result by luck. The standard threshold is 0.05. Below 0.05 = statistically significant.

In this project: Energy sector model: $p = 0.034$ (significant). Global model: $p = 0.090$ (not significant).

LLR p-value (Log-Likelihood Ratio)

Tests whether your entire logistic regression model — all predictors combined — explains the data significantly better than always predicting the majority class. It compares the full model against a model with no predictors at all.

In this project: This is the first number to check. If LLR $p > 0.05$, the model as a whole is not working.

McFadden Pseudo R^2

A measure of how much of the variation in outcomes your model explains. McFadden R^2 of 0.10-0.20 is considered good for logistic regression. Above 0.20 is excellent. Unlike linear R^2 , values above 0.30 are rare in stock prediction.

In this project: Your Energy model: 0.180. Your Utilities model: 0.218 (best). Global model: 0.011 (very weak).

AUC-ROC (Area Under the Curve)

Measures how well the model separates outperformers from underperformers. Ranges from 0.5 (random guessing — coin flip) to 1.0 (perfect). If you randomly pick one outperformer and one underperformer, AUC is the probability your model scores the outperformer higher.

In this project: Utilities model: AUC = 0.805 — very strong. Energy: 0.771. Global model: 0.555 — barely above chance.

Wald Test / z-statistic

Tests whether a single predictor (one ratio) contributes significantly to the model on its own. Each variable gets its own z-score and p-value. If $p < 0.05$ for a variable, that ratio significantly predicts outperformance after controlling for all other ratios.

In this project: In Energy: ROA and asset turnover are individually significant. In Consumer Staples: net income growth.

Odds Ratio

The exponentiated logistic regression coefficient. An odds ratio of 1.5 for ROA means a one-unit increase in standardized ROA multiplies the odds of outperforming by 1.5x. An odds ratio below 1.0 means the variable is negatively associated with outperformance.

In this project: More interpretable than raw coefficients. Always reported alongside p-values in publication tables.

95% Confidence Interval

A range within which the true coefficient is likely to fall 95% of the time if the study were repeated. A narrow interval means a precise estimate. A CI that includes 1.0 for an odds ratio means the result is not statistically significant.

In this project: Always reported alongside odds ratios in published logistic regression tables.

VIF (Variance Inflation Factor)

Measures how much a predictor variable overlaps with the other predictors — called multicollinearity. $VIF = 1$ means no overlap. $VIF > 10$ means severe overlap — the variable should be dropped because it provides redundant information.

In this project: Your result: all 13 ratios had VIF below 6. No multicollinearity — a clean result.

Silhouette Score

Measures how well-separated clusters are in K-Means analysis. Ranges from -1 (wrong clusters) to 0 (random) to 1 (perfect clusters). Your result of 0.179 indicates moderate but meaningful clustering.

In this project: Confirms that sectors have financially distinct profiles, justifying separate sector models.

Sharpe Ratio

Measures return earned per unit of risk taken. Formula: (Portfolio Return minus Risk-Free Rate) divided by Standard Deviation of Returns. Higher Sharpe ratio = better risk-adjusted performance. Used to compare portfolios that have different levels of volatility.

In this project: Used in backtesting to compare your model portfolio against SPY on a risk-adjusted basis.

Confusion Matrix

A 2x2 table showing: how many outperformers the model correctly identified (True Positives), how many underperformers correctly identified (True Negatives), and how many it got wrong in each direction (False Positives and False Negatives).

In this project: From this table you derive Accuracy, Precision, Recall, and F1-score — all reported in results.

SMOTE (Synthetic Minority Oversampling)

A technique to handle class imbalance. If outperformers are rarer than underperformers, SMOTE creates synthetic new outperformer examples by blending existing ones — so the model trains on a more balanced dataset.

In this project: Your class split is 46.6% vs 53.4% — mild imbalance. SMOTE applied as best practice for rigor.

Paired t-test

Tests whether two sets of measurements are significantly different from each other. In backtesting, you compare the portfolio's quarterly returns against SPY's quarterly returns — quarter by quarter — and test whether the difference is statistically significant.

In this project: If the paired t-test gives $p < 0.05$, your portfolio outperforms SPY not by luck but by design.

PART 4 — FINANCIAL STATEMENT TERMS

These are the raw line items from company financial statements used to compute the ratios.

Term	Plain-English Meaning
Revenue	Total sales — all money the company received from customers
Cost of Revenue	Direct cost of producing the goods or services sold
Gross Profit	Revenue minus Cost of Revenue
Operating Income	Gross Profit minus all operating expenses (salaries, rent, marketing)
Net Income	Final profit after all costs, interest, and taxes are paid
Total Assets	Everything the company owns — cash, buildings, equipment, inventory
Total Liabilities	Everything the company owes — loans, payables, long-term debt
Total Equity	Assets minus Liabilities — what shareholders actually own
Current Assets	Assets convertible to cash within 12 months
Current Liabilities	Debts due within 12 months
Long-Term Debt	Loans and bonds due more than 12 months from now
Operating Cash Flow	Actual cash generated from running the core business
Capital Expenditures	Cash spent on physical assets — equipment, buildings, infrastructure
Free Cash Flow	Operating Cash Flow minus Capital Expenditures
Retained Earnings	Cumulative profits kept in the company rather than paid as dividends
Shares (Basic)	Number of common shares outstanding — used to compute market cap
Market Cap	Price per share times shares outstanding — total company value

PART 5 — THE LOGISTIC REGRESSION MODEL EXPLAINED

5.1 What It Predicts

Logistic regression does not predict a stock's price. It predicts a YES or NO answer: will this stock beat the market next quarter? The output is always a probability between 0 and 1. Above 0.5 = predicted outperformer. Below 0.5 = predicted underperformer.

5.2 The Formula

Step 1 — Compute the linear score z :

$$z = \beta_{\text{intercept}} + \beta_{\text{gross_margin}}(\text{gross_margin}) + \beta_{\text{operating_margin}}(\text{operating_margin}) + \beta_{\text{net_margin}}(\text{net_margin}) + \beta_{\text{ROA}}(\text{ROA}) + \beta_{\text{ROE}}(\text{ROE}) + \beta_{\text{current_ratio}}(\text{current_ratio}) + \beta_{\text{debt_to_equity}}(\text{debt_to_equity}) + \beta_{\text{asset_turnover}}(\text{asset_turnover}) + \beta_{\text{fcf_margin}}(\text{fcf_margin}) + \beta_{\text{cash_ratio}}(\text{cash_ratio}) + \beta_{\text{revenue_growth}}(\text{revenue_growth}) + \beta_{\text{ni_growth}}(\text{ni_growth}) + \beta_{\text{capex_ratio}}(\text{capex_ratio})$$

Step 2 — Convert z to a probability:

$$P(\text{outperformer}) = 1 \div (1 + e^{(-z)})$$

5.3 What the β Coefficients Mean

Each β is a weight the model assigns to one ratio. The model learns these weights from historical data — finding the combination that best separates past outperformers from underperformers. Positive β = higher ratio values → higher outperformer probability. Negative β = the opposite.

5.4 Why Sector-Specific Models

A global model tries to apply the same weights to Apple and ExxonMobil simultaneously. But what makes an energy company outperform (asset efficiency, ROA) is completely different from what makes a consumer staples company outperform (earnings momentum). Separate models per sector allow each to learn its own weights — producing significantly stronger results.

5.5 Preliminary Results Summary

Model	LLR p-value	R ²	AUC	Verdict
Global — all sectors	0.090	0.011	0.555	Not significant
Energy	0.034	0.180	0.771	SIGNIFICANT
Industrials	0.028	0.140	0.718	SIGNIFICANT
Consumer Staples	0.030	0.136	0.735	SIGNIFICANT
Real Estate	0.035	0.178	0.754	SIGNIFICANT
Utilities	0.059	0.218	0.805	Borderline
IT, Healthcare, etc.	>0.15	<0.09	<0.68	Not significant

This guide covers every term, ratio, and measure you will encounter throughout this project. Keep it as a reference during the analysis — from data preparation through to the final portfolio performance evaluation. Each concept connects directly to a specific step in the methodology.